Beria Ayşenur Can

Particle Physicist| Data Scientist & Engineer

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Profile

- Graduate researcher at ITU and Member of the CMS Collaboration at CERN. Focused on detector physics, simulation optimization, and the analysis of high-energy collision data. Experienced in reconstruction algorithms and statistical methods to study fundamental interactions and assess detector performance.

Education

- **Istanbul Technical University** *2024 – Present*
  M.Sc. in Physics Engineering Istanbul, Türkiye

  Specialization in particle physics & computational physics

- **Istanbul Technical University** *2020 – 2024*
  B.Sc. in Physics Engineering Istanbul, Türkiye

  Graduated **3rd in department**; GPA: 3.3/4.0
  Thesis: **Investigation of Radon Particle Formation and Measurement for Earthquake Prediction Using Deep Learning**
  Focus on computational physics, machine learning, simulation techniques.

Experience

- **Adin.ai** *2025 – Present*
  Data Engineer & Scientist Istanbul, Türkiye

  – Developing and deploying scalable **REST APIs** using **FastAPI** to provide real-time marketing analytics.
  – Implementing complex **statistical research** and metric calculations (**CPM, CPR, Nielsen**) using **NumPy** and **Pandas** to evaluate advertising performance.
  – Building and maintaining robust **data pipelines** using **SQL (PostgreSQL/DBeaver)** and **Python**, ensuring seamless data flow and **visualization**.
  – Utilizing **Postman** for API testing and integrating **cloud-based workflows** within **AWS** environments (**Cognito, S3**) for high-concurrency requests.

- **SAS** *2024*
  Associate Analytical Consultant / Technical Intern Istanbul, Türkiye

  – Developed **Machine Learning** models for **Fraud Detection** using **Python, SQL,** and **Scikit-learn**.
  – Performed **anomaly detection** and predictive analytics to identify fraudulent patterns, enhancing financial security protocols.
  – Conducted **model validation** and result interpretation, translating complex analytical findings into actionable business insights.

- **TUBITAK 1001 Project** *2023 – 2024*
  Research Student (Scholar) Istanbul, Türkiye

  – Analyzed radon concentration data; applied **time-series modeling** and **Deep Learning** to investigate seismic correlations.
  – Managed the **end-to-end data pipeline**, including acquisition, **noise reduction**, and visualization of environmental sensor data.

Technical Skills

- **Programming:** Python, C++, SQL, Matlab, Bash
- **Machine Learning & Data Science:** Scikit-learn, NumPy, Pandas, SciPy, Anomaly Detection, Time-Series Forecasting
- **Data Engineering & Web:** FastAPI, REST APIs, ETL Pipelines, DBeaver, Postman
- **Physics & Analytics:** ROOT (CERN), COMSOL, SimFlow
- **Cloud & Tools:** AWS (Cognito/S3/EC2), Linux, Git, Docker, Matplotlib/Seaborn (Visualization)